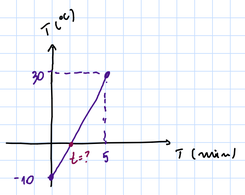
EX 06



$y = ax + b$

lei DE FORMAÇÃO

$T(0) = -10°C$

$T(5) = 30°C$

$T(t) = a.t + b$

$$\begin{cases} \cancel{a.0} + b = -10 \\ a.5 + b = 30 \end{cases}$$

$\boxed{b = -10}$

$5a + b = 30$

$5a = 40$

$\boxed{a = 8}$

$\boxed{T(t) = 8.t - 10}$

$8t - 10 = 0$

$8t = 10$

$t = 10/8$

$\boxed{t = 1{,}25 \text{ min}}$

$a > 0$ CRESCENTE

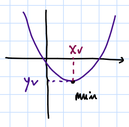
$a < 0$ DECRESCENTE

EX15 $K = ?$

$a = 1 \quad b = -K \quad c = 4$

$f(x) = x^2 - Kx + 4$

$-\frac{9}{4}$ mínimo



$\boxed{y_v = -\frac{9}{4}}$

$y_v = -\frac{\Delta}{4a} = -\frac{9}{4}$

$-\frac{(b^2 - 4ac)}{4a} = -\frac{9}{4}$

$\cancel{-}\frac{(-K)^2 - 4.1.4)}{\cancel{4.1}} = \cancel{-}\frac{9}{\cancel{4}}$

$K^2 - 16 = 9$

$K^2 = 25$

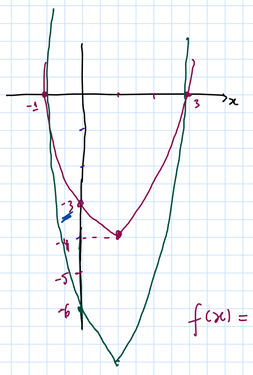
$\boxed{K = \pm 5}$

c) $a > 0$

$x \leq x_v$ DECRESCENTE

$x \geq x_v$ CRESCENTE

EX20



$x = -1 \longrightarrow x + 1 = 0$

$x = 3 \longrightarrow x - 3 = 0$

$f(x) = (x+1).(x-3)$

$f(x) = x^2 - 3x + x - 3$

$f(x) = x^2 - 2x - 3$

$f(x) = ax^2 + bx + c$

TRANSLAÇÃO $f(x) = 2x^2 - 4x - 6 = 2.(x^2 - 2x + 3)$

EX 27

$f(x) = x^2 - 5x + 6$

$g(x) = x + 4$

$f(g(x)) = 0 \qquad f \circ g(x) = 0$

$f(x) = x^2 - 5x + 6$

$f(g(x)) = g(x)^2 - 5.g(x) + 6$

$f(g(x)) = (x+4)^2 - 5.(x+4) + 6$

$f(g(x)) = x^2 + 8x + 16 - 5x - 20 + 6$

$f(g(x)) = x^2 + 3x + 2 = 0$

$x = \frac{-3 \pm \sqrt{9 - 8}}{2}$

$x = \frac{-3 \pm 1}{2}$ → $x_1 = -2$, $x_2 = -1$

$(a+b)^2 = a^2 + 2ab + b^2$

	a	b
a	a^2	ab
b	ab	b^2

EXERCÍCIO 14 d)

$f(x) = 4x^2 - 3x + 2$

$g(h) = \frac{f(a+h) - f(a)}{h} \longrightarrow g(h) = \frac{\cancel{4a^2} + 8ah + 4h^2 \cancel{-3a} - 3h \cancel{+2} - (\cancel{4a^2} - 3a - 2)}{h}$ $\quad +3a \quad -2$

$f(a) = 4a^2 - 3a + 2$

$f(a+h) = 4.(a+h)^2 - 3.(a+h) + 2$

$f(a+h) = 4.(a^2 + 2ah + h^2 - 3a - 3h + 2$

$f(a+h) = 4a^2 + 8ah + 4h^2 - 3a - 3h + 2$

$g(h) = \frac{8ah + 4h^2 - 3h}{h}$

$g(h) = \frac{\cancel{h}.(8a + 4h - 3)}{\cancel{h}}$

$\boxed{g(h) = 8a + 4h - 3}$

$D = \mathbb{R}$

EX 36

$f(x) = \frac{-x^2 + 8x - 16}{3}$ → $D = [4, +\infty)$

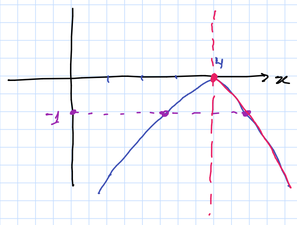
MOSTRE QUE $f^{-1}(x) = 4 + \sqrt{-3x}$

domínio $f^{-1}(x)$

$x = \frac{-8 \pm \sqrt{64 - 64}}{-2}$

$x = \frac{-8 \pm 0}{-2}$

$x = 4$



$x_1 \neq x_2 \longrightarrow f(x_1) = f(x_2)$

$D = \{x \in \mathbb{R} / x \geq 4\} = [4, +\infty)$

$Im = \{y \in \mathbb{R} / y \leq 0 = (-\infty, 0]$

INVERSA

$D = \{x \in \mathbb{R} / x \leq 0\} = (-\infty, 0]$

$Im = \{y \in \mathbb{R} / y \geq 4\} = [4, +\infty)$

$y = \frac{-x^2 + 8x - 16}{3}$

$x = \frac{-y^2 + 8y - 16}{3}$ $\qquad y^2 - 8y + 16 = (y-4)^2$

$3x = -(y^2 - 8y + 16)$

$3x = -(y-4)^2$

$-3x = (y-4)^2$

$y - 4 = \pm\sqrt{-3x}$

$y = 4 \pm \sqrt{-3x}$ INVERSA?

$f^{-1}(x) = 4 \pm \sqrt{-3x}$

$\boxed{f^{-1}(x) = 4 + \sqrt{-3x}}$ → função INVERSA

$\cancel{f^{-1}(x) = 4 - \sqrt{-3x}}$

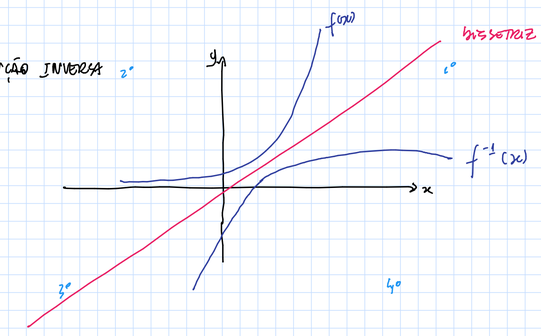
$x = -2$

$f^{-1}(-2) = 4 + \sqrt{-3.-2} \approx 6{,}45$

$x = -2$

$f^{-1}(-2) = 4 - \sqrt{-3.-2} = 4 - 2{,}45 \approx 1{,}55$

FUNÇÃO INVERSA

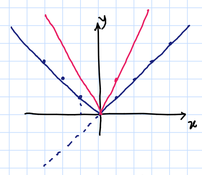


TRANSLAÇÃO

FUNÇÃO MODULAR

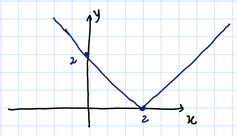
$f(x) = |x|$

$I = [0, +\infty)$



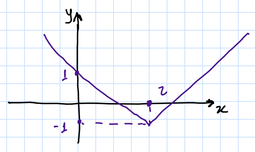
$f(x) = |x - 2|$

$Im = [0, +\infty)$



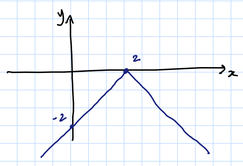
$f(x) = |x - 2| - 1$

$Im = [-1, \infty)$



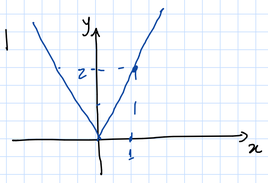
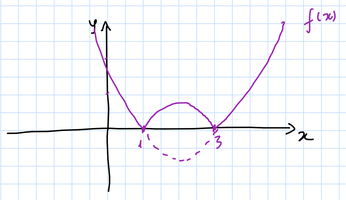
$f(x) = -|x - 2|$

$Im = (-\infty, 0]$



$f(x) = |2x|$

$Im = [0, +\infty)$



$f(x) = |x^2 - 4x + 3|$